

LILI JU

CURRICULUM VITAE

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EDUCATION

- Ph.D.** in *Applied Mathematics*, 2002/05, Iowa State University
- M.S.** in *Computational Mathematics*, 1998/07, Chinese Academy of Sciences
- B.S.** in *Mathematics*, 1995/07, Wuhan University, China

WORKING EXPERIENCES

- 2013/01 – Present Professor
Department of Mathematics, University of South Carolina
- 2008/08 – 2012/12 Associate Professor
Department of Mathematics, University of South Carolina
- 2004/08 – 2008/08 Assistant Professor
Department of Mathematics, University of South Carolina
- 2002/09 – 2004/08 Industrial Postdoctoral Fellow
Institute for Mathematics and its Applications, University of Minnesota
- 1998/08 – 2002/08 Graduate Teaching/Research Assistant
Department of Mathematics and Computing Center, Iowa State University

AWARDS AND HONORS

- *SIAM Fellow*, Society for Industrial and Applied Mathematics, 2025.
- *World's Top 2% Scientists* by Stanford/Elsevier, 2023 – 2025.
- *Plenary Speaker*, The 45th SIAM Southeastern Atlantic Section Annual Meeting, 2023.
- *Plenary Speaker*, The 12th China National Conference on Computational Mathematics, 2019.
- *ACM Gordon Bell Prize finalist* in High Performance Computing, 2016.
- *Featured Scholar of June 2012*, University of South Carolina, 2012.

PROFESSIONAL ACTIVITIES

- Member of the Committee on Committees and Appointments, Society for Industrial and Applied Mathematics, 2024/01– Present
- President (2008-2009), Secretary-Treasurer (2007-2008), SIAM Southeastern Atlantic Section

EDITORSHIP

- Editor-in-Chief (2026/01 – Present), Associate Editor (2021/10 – 2025/12), *Numerical Methods for Partial Differential Equations*
- Associate Editor (2024/02 – Present), *Mathematics of Computation*
- Editorial Board Member (2023/01 – Present), *Journal of Scientific Computing*
- Associate Editor (2022/07 – Present), *Advances in Computational Science and Engineering*
- Associate Editor (2021/07 – Present), *Advances in Applied Mathematics and Mechanics*
- Associate Editor (2021/01 – Present), *Numerical Math.: Theory, Methods and Applications*
- Associate Editor (2021/06 – 2023/09), *Advances in Continuous and Discrete Models*
- Associate Editor (2012/01 – 2017/12), *SIAM Journal on Numerical Analysis*

RESEARCH INTERESTS

Numerical algorithms and analysis
Scientific machine learning
Computational geoscience
Imaging and computer vision
High performance computing

GRANT SUPPORT

- Department of Energy, DE-SC0025527, “*DyGenAI: Dynamic Generative Artificial Intelligence for Prediction and Control of High-Dimensional Nonlinear Complex Systems*”, PI, 09/01/2024-08/31/2027, \$450,000.
- National Science Foundation, DMS-2409634, “*Structure-Preserving Linear Schemes for the Diffuse-Interface Models of Incompressible Two-Phase Flows with Matched Densities*”, PI, 08/01/2024-07/31/2027, \$294,473.
- Department of Energy, DE-SC0022254, “*Reliable and Efficient Machine Learning for Leadership Facility Scientific Data Analytics*”, PI, 09/01/2021-08/31/2025, \$300,000.
- National Science Foundation, DMS-2109633, “*Maximum Bound Principle-Preserving Time Integration Methods for Some Semilinear Parabolic Equations*”, PI, 07/01/2021-06/30/2024, \$150,000.
- Department of Energy, DE-SC0020270, “*Efficient and Scalable Time-Stepping Algorithms and Reduced-Order Modeling for Ocean System Simulations*”, PI, 09/01/2019-08/031/2022, \$375,000.
- National Science Foundation, DMS-1818438, “*Study on Localized Exponential Time Differencing Methods for Evolution Partial Differential Equations*”, PI, 08/01/2018-07/31/2021, \$150,000.
- National Science Foundation, DMS-1748357, “*The Ninth Annual Graduate Student Miniconference in Computational Mathematics*”, Co-PI (PI: Zhu Wang), 01/01/2018-12/31/2018, \$7,280. (**Conference grant**)
- Department of Energy, DE-SC0016540, “*Grid Generation, Coupling Strategies, and Spatially-Dependent Time Stepping for Ocean Tidal/Estuary Systems and Other ESM Components*”, PI, 09/01/2016-08/31/2019, \$584,495.
- National Science Foundation, DMS-1521965, “*Fast and Stable Compact Exponential Time Difference Based Methods for Some Parabolic Equations*”, PI, 08/01/2015-07/31/2018, \$200,954.
- National Science Foundation, DMS-1215659, “*Numerical Improvements, Mesh Adaptation and Parameter Identification for Parallel Finite Element Stokes Ice Sheet Modeling*”, PI, 08/01/2012-07/31/2015, \$157,618.
- Department of Energy, DE-SC0008087-ER65393, “*Predicting Ice Sheet and Climate Evolution at Extreme Scales (PISCEES)*”, PI, 06/15/2012-06/14/2017, \$348,420.
- National Science Foundation, DMS-0913491, “*Study on Algorithms and Applications of Centroidal Voronoi Tessellations*”, PI, 09/01/2009-08/31/2012, \$180,000.
- Society for Industrial and Applied Mathematics, “*The 33rd SIAM Southeastern-Atlantic Section Annual Meeting*”, PIs (L. Szekely and L. Ju), 2009, \$4,180. (**Conference grant**)
- Department of Energy, DE-FG02-07ER64431, “*New Grid and Discretization Technologies for Ocean and Ice Simulations*”, PI, 07/15/2007-07/14/2011, \$236,871.
- National Science Foundation, DMS-0609575, “*Some Problems on Analyses and Applications of Centroidal Voronoi Tessellations*”, PI, 08/01/2006-07/31/2009, \$122,781.
- South Carolina Research Foundation, RPS-13060-06-12306, “*Some Applications of Centroidal Voronoi Tessellations*”, PI, 04/01/2006-06/30/2007, \$5,000.

PUBLICATIONS

DISSERTATION AND THESIS

- “*Probabilistic and Parallel Algorithms for Centroidal Voronoi Tessellations with Application to Meshless Computing and Numerical Analysis on Surface*”, **Ph.D. Dissertation**, Iowa State University, 2002.
- “*Parallel Implementation of a Pressure-Correction Projection Method for the Unsteady Incompressible Navier-Stokes Equations*”, **M.S. Thesis**, Institute of Computational Mathematics and Scientific/Engineering Computing, Chinese Academy of Sciences, 1998.

RESEARCH PAPERS

Journal articles

1. L. JU AND L.-B. ZHANG, Parallel implementation of a pressure-correction projection method for the unsteady incompressible Navier-Stokes equations, *Mathematica Numerica Sinica*, Vol. **20**, pp. 325-336, 1998.
2. Q. DU, M. GUNZBURGER AND L. JU, Meshfree, probabilistic determination of point sets and support regions for meshless computing, *Computer Methods in Applied Mechanics and Engineering*, Vol. **191**, pp. 1349-1366, 2002.
3. L. JU, Q. DU AND M. GUNZBURGER, Probabilistic methods for centroidal Voronoi tessellations and their parallel implementations, *Parallel Computing*, Vol. **28**, pp. 1477-1500, 2002.
4. Q. DU, M. GUNZBURGER AND L. JU, Constrained centroidal Voronoi tessellations for surfaces, *SIAM Journal on Scientific Computing*, Vol. **24**, pp. 1488-1506, 2003.
5. G. LUECKE, M. KRAEVA AND L. JU, Comparing the performance of MPICH with Cray’s MPI and with SGI’s MPI, *Concurrency and Computation: Practice and Experience*, Vol. **15**, pp. 779-802, 2003.
6. Q. DU, M. GUNZBURGER AND L. JU, Voronoi-based finite volume methods, optimal Voronoi meshes, and PDEs on the sphere, *Computer Methods in Applied Mechanics and Engineering*, Vol. **192**, pp. 3933-3957, 2003.
7. M. GUNZBURGER, L.S. HOU AND L. JU, Approximation of exact boundary controllability problems for the 1-D wave equation by optimization-based methods, *Contemporary Mathematics*, Vol. **330**, pp. 133-153, 2003.
8. Q. DU AND L. JU, Numerical simulation of the quantized vortices in a thin superconducting hollow sphere, *Journal of Computational Physics*, Vol. **201**, pp. 511-530, 2004.
9. Q. DU AND L. JU, Approximations of a Ginzburg-Landau model for superconducting hollow spheres based on spherical centroidal Voronoi tessellations, *Mathematics of Computation*, Vol. **74**, pp. 1257-1280, 2005.
10. Q. DU AND L. JU, Finite volume methods on spheres and spherical centroidal Voronoi meshes, *SIAM Journal on Numerical Analysis*, Vol. **43**, pp. 1673-1692, 2005.
11. L. JU, M. HURDAL, K. REHM, K. SCHAPER, J. STERN AND D. ROTTENBERG, Quantitative evaluation of three cortical surface flattening methods, *Neuroimage*, Vol. **28**, pp. 869-880, 2005.
12. Q. DU, M. EMELIANENKO AND L. JU, Convergence of the Lloyd algorithm for computing centroidal Voronoi tessellations, *SIAM Journal on Numerical Analysis*, Vol. **44**, pp. 102-119, 2006.
13. Q. DU, M. GUNZBURGER, L. JU AND X. WANG, Centroidal Voronoi tessellation algorithms for image compression, segmentation, and multichannel restoration, *Journal of Mathematical Imaging and Vision*, Vol. **24**, pp. 177-194, 2006.

14. M. GUNZBURGER, L.S. HOU AND L. JU, A numerical method for exact boundary controllability problems for the 2-D wave equation, *Computers & Mathematics with Applications*, Vol. **51**, pp. 721-750, 2006.
15. L. JU, M. GUNZBURGER AND W. ZHAO, Adaptive finite element methods for elliptic PDEs based on conforming centroidal Voronoi Delaunay triangulations, *SIAM Journal on Scientific Computing*, Vol. **28**, pp. 2023-2053, 2006.
16. L. JU, Conforming centroidal Voronoi Delaunay triangulation for quality mesh generation, *International Journal of Numerical Analysis and Modeling*, Vol. **4**, pp. 531-547, 2007.
17. M. EMELIANENKO, L. JU AND A. RAND, Nondegeneracy and weak global convergence of the Lloyd algorithm in $\mathbb{R}^d$, *SIAM Journal on Numerical Analysis*, Vol. **46**, pp. 1423-1441, 2008.
18. W. ZHAO, L. TIAN AND L. JU, Convergence analysis of a splitting scheme for stochastic differential equations, *International Journal of Numerical Analysis and Modeling*, Vol. **5**, pp. 673-692, 2008.
19. L. JU, H.-C. LEE AND L. TIAN, Numerical simulations of the steady Navier-Stokes equations using adaptive meshing schemes, *International Journal for Numerical Methods in Fluids*, Vol. **56**, pp. 703-721, 2008.
20. T. RINGLER, L. JU AND M. GUNZBURGER, A multi-resolution method for climate system modeling, *Ocean Dynamics*, Vol. **58**, pp. 475-498, 2008.
21. H. NGUYEN, J. BURKARDT, M. GUNZBURGER, L. JU AND Y. SAKA, Constrained CVT meshes and a comparison of triangular mesh generators, *Computational Geometry: Theory and Applications*, Vol. **42**, pp. 1-19, 2009.
22. L. JU AND Q. DU, A finite volume method on general surfaces and its error estimates, *Journal of Mathematical Analysis and Applications*, Vol. **352**, pp. 645-668, 2009.
23. Q. DU, L. JU AND L. TIAN, Analysis of a mixed finite-volume discretization for fourth-order equations on general surfaces, *IMA Journal of Numerical Analysis*, Vol. **29**, pp. 376-403, 2009.
24. L. JU, W. WU AND W. ZHAO, Adaptive finite volume methods for steady convection-diffusion equations with mesh optimization, *Discrete and Continuous Dynamical Systems B*, Vol. **11**, pp. 669-690, 2009.
25. H. NGUYEN, M. GUNZBURGER, L. JU AND J. BURKARDT, Adaptive anisotropic meshing for steady convection-dominated problems, *Computer Methods in Applied Mechanics and Engineering*, Vol. **198**, pp. 2964-2981, 2009.
26. J. WANG, L. JU AND X. WANG, An edge-weighted centroidal Voronoi tessellation model for image segmentation, *IEEE Transactions on Image Processing*, Vol. **18**, pp. 1844-1858, 2009.
27. L. JU, L. TIAN AND D. WANG, A posteriori error estimates for finite volume approximations of elliptic equations on general surfaces, *Computer Methods in Applied Mechanics and Engineering*, Vol. **198**, pp. 716-726, 2009.
28. L. ZHU, Y. WANG, L. JU AND D. WANG, A variational phase field method for curve smoothing, *Journal of Computational Physics*, Vol. **229**, pp. 2390-2400, 2010.
29. Q. DU, M. GUNZBURGER AND L. JU, Advances in studies and applications of centroidal Voronoi tessellations, *Numerical Mathematics: Theory, Methods and Applications*, Vol. **3**, pp. 119-142, 2010.
30. E. CHOATE, M. FOREST AND L. JU, Effects of strong anchoring on the dynamic moduli of heterogeneous nematic polymers II: Oblique anchoring angles, *Rheologica Acta*, Vol. **49**, pp. 335-347, 2010.
31. W. ZHAO, G. ZHANG AND L. JU, A stable multi-step scheme for solving backward stochastic differential equations, *SIAM Journal on Numerical Analysis*, Vol. **48**, pp. 1369-1394, 2010.
32. H. ZHANG, L. JU, M. GUNZBURGER, T. RINGLER AND S. PRICE, Coupled models and parallel simulations for three dimensional full-Stokes ice sheet modeling, *Numerical Mathematics: Theory, Methods and Applications*, Vol. **4**, pp. 359-381, 2011.

33. Q. DU, L. JU AND L. TIAN, Finite element approximation of the Cahn-Hilliard equations on surfaces, *Computer Methods in Applied Mechanics and Engineering*, Vol. **200**, pp. 2458-2470, 2011.
34. J. WANG, L. JU AND X. WANG, Image segmentation using local variation and edge-weighted centroidal Voronoi tessellations, *IEEE Transactions on Image Processing*, Vol. **20**, pp. 3242-3256, 2011.
35. T. RINGLER, D. JACOBSEN, M. GUNZBURGER, L. JU, M. DUDA AND W. SKAMAROCK, Exploring a multi-resolution modeling approach within the shallow-water equations, *Monthly Weather Review*, Vol. **139**, pp. 3348-3368, 2011.
36. W. LENG, L. JU, M. GUNZBURGER, S. PRICE AND T. RINGLER, A parallel high-order accurate finite element nonlinear Stokes ice sheet model and benchmark experiments, *Journal of Geophysical Research*, Vol. **117**, F01001 (24 pages), 2012.
37. L. JU, L. TIAN, X. XIAO AND W. ZHAO, Covolume-upwind finite volume approximations for linear elliptic partial differential equations, *Journal of Computational Physics*, Vol. **231**, pp. 6097-6120, 2012.
38. Y. WANG, L. JU, D. WANG AND X. WANG, Generalized edge-weighted centroidal Voronoi tessellations for geometry processing, *Computers & Mathematics with Applications*, Vol. **64**, pp. 2663-2681, 2012.
39. W. LENG, L. JU, M. GUNZBURGER AND S. PRICE, Manufactured solutions and the verification of three-dimensional Stokes ice-sheet models, *The Cryosphere*, Vol. **7**, pp. 19-29, 2013.
40. W. ZHAO, Y. LI AND L. JU, Error estimates of the Crank-Nicolson scheme for solving backward stochastic differential equations, *International Journal of Numerical Analysis and Modeling*, Vol. **10**, pp. 876-898, 2013.
41. Q. DU, L. JU, L. TIAN AND K. ZHOU, A posteriori error analysis of finite element method for linear nonlocal diffusion and peridynamic models, *Mathematics of Computation*, Vol. **82**, pp. 1889-1922, 2013.
42. Y. CAO, L. JU, Y. ZHOU AND S. WANG, 3D superalloy grain segmentation using a multi-channel edge-weighted centroidal Voronoi tessellation algorithm, *IEEE Transactions on Image Processing*, Vol. **22**, pp. 4123-4135, 2013.
43. W. ZHAO, W. ZHANG AND L. JU, A numerical method and its error estimates for the decoupled forward-backward stochastic differential equations, *Communications in Computational Physics*, Vol. **15**, pp. 618-646, 2014.
44. W. LENG, L. JU, Y. XIE, T. CUI AND M. GUNZBURGER, Finite element three-dimensional Stokes ice sheet dynamics model with enhanced local mass conservation, *Journal of Computational Physics*, Vol. **274**, pp. 299-311, 2014.
45. L. JU, X. LIU, AND W. LENG, Compact implicit integration factor methods for a family of semilinear fourth-order parabolic equations, *Discrete and Continuous Dynamical Systems B*, Vol. **19**, pp. 1667-1687, 2014.
46. W. LENG, L. JU, M. GUNZBURGER AND S. PRICE, A parallel computational model for three-dimensional, thermo-mechanical Stokes flow simulations of glaciers and ice sheets, *Communications in Computational Physics*, Vol. **16**, pp. 1056-1080, 2014.
47. L. JU, J. ZHANG, L. ZHU AND Q. DU, Fast explicit integration factor methods for semilinear parabolic equations, *Journal of Scientific Computing*, Vol. **62**, pp. 431-455, 2015.
48. H. TIAN, L. JU AND Q. DU, Nonlocal convection-diffusion problems and finite element approximations, *Computer Methods in Applied Mechanics and Engineering*, Vol. **289**, pp. 60-78, 2015.
49. Y. ZHOU, L. JU AND S. WANG, Multiscale superpixels and supervoxels based on hierarchical edge-Weighted centroidal Voronoi tessellation, *IEEE Transactions on Image Processing*, Vol. **24**, pp. 3834-3845, 2015.

50. T. ZHANG, L. JU, W. LENG, S. PRICE AND M. GUNZBURGER, Thermomechanically coupled glacier modeling for land-terminating glaciers: A comparison of two-dimensional, first-order and three-dimensional, full Stokes approaches, *Journal of Glaciology*, Vol. **61**, pp. 702-712, 2015.
51. L. JU, J. ZHANG AND Q. DU, Fast and accurate algorithms for simulating coarsening dynamics of Cahn-Hilliard equations, *Computational Materials Science*, Vol. **108**, pp. 272-282, 2015.
52. W. ZHAO, W. ZHANG AND L. JU, A multistep scheme for decoupled forward-backward stochastic differential equations, *Numerical Mathematics: Theory, Methods and Applications*, Vol. **9**, pp. 262-288, 2016.
53. X. ZHANG, M. GUNZBURGER AND L. JU, Nodal-type collocation method for hypersingular integral equations and nonlocal diffusion problems, *Computer Methods in Applied Mechanics and Engineering*, Vol. **299**, pp. 401-420, 2016.
54. X. ZHANG, M. GUNZBURGER AND L. JU, Quadrature rules for finite element approximations of 1D nonlocal problems, *Journal of Computational Physics*, Vol. **310**, pp. 213-236, 2016.
55. X. WANG, L. JU AND Q. DU, Efficient and stable exponential time differencing Runge-Kutta methods for phase field elastic bending energy models, *Journal of Computational Physics*, Vol. **316**, pp. 21-38, 2016.
56. L. ZHU, L. JU AND W. ZHAO, Fast high-order compact exponential time differencing Runge-Kutta methods for second-order semilinear parabolic equations, *Journal of Scientific Computing*, Vol. **67**, pp. 1043-1065, 2016.
57. X. FAN, L. JU, X. WANG AND S. WANG, A fuzzy edge-weighted centroidal Voronoi tessellation model for image segmentation, *Computers & Mathematics with Applications*, Vol. **71**, pp. 2272-2284, 2016.
58. X. YANG AND L. JU, Efficient linear schemes with unconditionally energy stability for the phase field elastic bending energy model, *Computer Methods in Applied Mechanics and Engineering*, Vol. **315**, pp. 691-712, 2017.
59. T. ZHANG, S. PRICE, L. JU, W. LENG, J. BRONDEX, G. DURAND AND O. GAGLIARDINI, A comparison of two Stokes ice sheet models applied to the Marine Ice Sheet Model Intercomparison Project for plan view models (MISMIP3d), *The Cryosphere*, Vol. **11**, pp. 179-190, 2017.
60. X. YANG AND L. JU, Linear and unconditionally energy stable schemes for the binary fluid-surfactant phase field model, *Computer Methods in Applied Mechanics and Engineering*, Vol. **318**, pp. 1005-1029, 2017.
61. H. TIAN, L. JU AND Q. DU, A conservative nonlocal convection-diffusion model and asymptotically compatible finite difference discretization, *Computer Methods in Applied Mechanics and Engineering*, Vol. **320**, pp. 46-67, 2017.
62. L. JU AND Z. WANG, Exponential time differencing gauge method for incompressible viscous flows, *Communications in Computational Physics*, Vol. **22**, pp. 517-541, 2017.
63. L. JU, X. LI, Z. QIAO AND H. ZHANG, Energy stability and error estimates of exponential time differencing schemes for the epitaxial growth model without slope selection, *Mathematics of Computation*, Vol. **87**, pp. 1859-1885, 2018.
64. Q. DU, L. JU, X. LI AND Z. QIAO, Stabilized linear semi-implicit schemes for the nonlocal Cahn-Hilliard equation, *Journal of Computational Physics*, Vol. **363**, pp. 39-54, 2018.
65. S. DUO, L. JU AND Y. ZHANG, A fast algorithm for solving the space-time fractional diffusion equation, *Computers & Mathematics with Applications*, Vol. **75**, pp. 1929-1941, 2018.
66. H. LI, L. JU, C. ZHANG AND Q. PENG, Unconditionally energy stable linear schemes for a diffuse interface model with Peng-Robinson equation of state, *Journal of Scientific Computing*, Vol. **75**, pp. 993-1015, 2018.
67. S. LI, L. LUO, Z.J. WANG AND L. JU, An exponential time-integrator scheme for steady and unsteady inviscid flows, *Journal of Computational Physics*, Vol. **365**, pp. 206-225, 2018.

68. X. ZHANG, J. WU AND L. JU, An accurate and asymptotically compatible collocation method for nonlocal diffusion problems, *Applied Numerical Mathematics*, Vol. **133**, pp. 52-68, 2018.
69. Q. CHEN AND L. JU, Conservative finite volume schemes for solving the quasi-geostrophic equation on coastal-conforming unstructured primal-dual meshes, *Quarterly Journal of the Royal Meteorological Society*, Vol. **144**, pp. 1106-1122, 2018.
70. H. FENG, Y. GAO, L. JU, H. TIAN AND X. ZHANG, Nodal-type Newton-Cotes rules for fractional order hypersingular integrals, *East Asian Journal of Applied Mathematics*, Vol. **8**, pp. 697-714, 2018.
71. H. YANG, M. GUNZBURGER AND L. JU, Fast spherical centroidal Voronoi mesh generation: a Lloyd-preconditioned LBFGS method in parallel, *Journal of Computational Physics*, Vol. **367**, pp. 235-252, 2018.
72. T.-T.-P. HOANG, L. JU AND Z. WANG, Overlapping localized exponential time differencing methods for diffusion problems, *Communications in Mathematical Sciences*, Vol. **16**, pp. 1531-1555, 2018.
73. H. FENG, Y. GAO, L. JU AND X. ZHANG, A new collocation method for solving certain Hadamard finite-part integral equations, *International Journal of Numerical Analysis and Modeling*, Vol. **16**, pp. 240-254, 2019.
74. Q. DU, L. JU AND J. LU, A discontinuous Galerkin method for one-dimensional time-dependent nonlocal diffusion problems, *Mathematics of Computation*, Vol. **88**, pp. 123-147, 2019.
75. T.-T.-P. HOANG, W. LENG, L. JU, Z. WANG AND K. PIEPER, Conservative explicit local time-stepping schemes for the shallow water equations, *Journal of Computational Physics*, Vol. **382**, pp. 152-176, 2019.
76. Q. DU, L. JU AND J. LU, Analysis of fully discrete approximations for dissipative systems and application to time-dependent nonlocal diffusion problems, *Journal of Scientific Computing*, Vol. **78**, pp. 1438-1466, 2019.
77. J. HUANG, L. JU AND B. WU, A fast compact exponential time differencing method for semi-linear parabolic equations with Neumann boundary conditions, *Applied Mathematics Letters*, Vol. **94**, pp. 257-265, 2019.
78. Q. DU, L. JU, X. LI AND Z. QIAO, Maximum principle preserving exponential time differencing schemes for the nonlocal Allen-Cahn equation, *SIAM Journal on Numerical Analysis*, Vol. **57**, pp. 875-898, 2019.
79. C. ZHANG, H. LI, X. ZHANG AND L. JU, Linear and unconditionally energy stable schemes for the multi-component two-phase diffuse interface model with Peng-Robinson equation of state, *Communications in Computational Physics*, Vol. **26**, pp. 1071-1097, 2019.
80. W. LENG AND L. JU, An additive overlapping domain decomposition method for the Helmholtz equation, *SIAM Journal on Scientific Computing*, Vol. **41**, pp. A1252-A1277, 2019.
81. H. GAO, L. JU AND W. XIE, A stabilized semi-implicit Euler gauge-invariant method for the time-dependent Ginzburg-Landau equations, *Journal of Scientific Computing*, Vol. **80**, pp. 1083-1115, 2019.
82. X. LI, L. JU AND X.-C. MENG, Convergence analysis of exponential time differencing schemes for the Cahn-Hilliard equation, *Communications in Computational Physics*, Vol. **26**, pp. 1510-1529, 2019.
83. J. HUANG, L. JU AND B. WU, A fast compact time integrator method for a family of general order semilinear evolution equations, *Journal of Computational Physics*, Vol. **393**, pp. 313-336, 2019.
84. H. GAO, L. JU, R. DUDDU AND H. LI, An efficient second-order linear scheme for the phase field model of corrosive dissolution, *Journal of Computational and Applied Mathematics*, Vol. **367**, #112472 (16 pages), 2020.

85. H. TIAN, J. ZHANG AND L. JU, A spectral collocation method for nonlocal diffusion equations with volume constrained boundary conditions, *Applied Mathematics and Computation*, Vol. **370**, #124930 (15 pages), 2020.
86. Q. DU, L. JU, J. LU AND X. TIAN, A discontinuous Galerkin method with penalty for one-dimensional nonlocal diffusion problems, *Communications on Applied Mathematics and Computation*, Vol. **2**, pp. 31-55, 2020.
87. H. GAO, L. JU, X. LI AND R. DUDDU, A space-time adaptive finite element method with exponential time integrator for the phase field model of pitting corrosion, *Journal of Computational Physics*, Vol. **406**, #109191 (20 pages), 2020.
88. T.-T.-P. HOANG, L. JU AND Z. WANG, Nonoverlapping localized exponential time differencing methods for diffusion problems, *Journal of Scientific Computing*, Vol. **82**, #37 (27 pages), 2020.
89. T.-T.-P. HOANG, L. JU, W. LENG AND Z. WANG, High order explicit local time-stepping methods for hyperbolic conservation laws, *Mathematics of Computation*, Vol. **89**, pp. 1807-1842, 2020.
90. L. JU, W. LENG, Z. WANG AND S. YUAN, Numerical investigation of ensemble methods with block iterative solvers for evolution problems, *Discrete and Continuous Dynamical Systems B*, Vol. **25**, pp. 4905-4923, 2020.
91. X. MENG, T.-T.-P. HOANG, Z. WANG AND L. JU, Localized exponential time differencing methods for shallow water equations: algorithms and numerical study, *Communications on Computational Physics*, Vol. **29**, pp. 80-110, 2021.
92. Y. WANG, K. LUO, Z. CHEN, L. JU AND T. GUAN, DeepFusion: A simple way to improve traditional multi-view stereo methods using deep learning, *Knowledge-Based Systems*, Vol. **221**, #106968 (8 pages), 2021.
93. W. LENG AND L. JU, A diagonal sweeping domain decomposition method with source transfer for the Helmholtz equation, *Communications on Computational Physics*, Vol. **29**, pp. 357-398, 2021.
94. X. LI, L. JU AND T.-T.-P. HOANG, Overlapping domain decomposition-based exponential time differencing methods for semilinear parabolic equations, *BIT Numerical Mathematics*, Vol. **61**, pp. 1-36, 2021.
95. Q. DU, L. JU, X. LI AND Z. QIAO, Maximum bound principles for a class of semilinear parabolic equations and exponential time-differencing schemes, *SIAM Review*, Vol. **63**, pp. 317-359, 2021.
96. L. JU, X. LI, Z. QIAO AND J. YANG, Maximum bound principle preserving integrating factor Runge-Kutta methods for semilinear parabolic equations, *Journal of Computational Physics*, Vol. **439**, #110405 (18 pages), 2021.
97. J. LI, L. JU, Y. CAI AND X. FENG, Unconditionally maximum bound principle preserving linear schemes for the conservative Allen-Cahn equation with nonlocal constraint, *Journal of Scientific Computing*, Vol. **87**, #98 (32 pages), 2021.
98. J. LI, X. LI, L. JU AND X. FENG, Stabilized integrating factor Runge-Kutta method and unconditional preservation of maximum bound principle, *SIAM Journal on Scientific Computing*, Vol. **43**, pp. A1780-A1802, 2021.
99. A. GRUBER, M. GUNZBURGER, L. JU, Y. TENG AND Z. WANG, Nonlinear level set learning for function approximation on sparse data with applications to parametric differential equations, *Numerical Mathematics: Theory, Methods and Applications*, Vol. **14**, pp. 839-861, 2021.
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218. Y. TENG, X. ZHANG, Z. WANG AND L. JU, Learning Green's functions of linear reaction-diffusion equations with application to fast numerical solver, *Proceedings of Third Mathematical and Scientific Machine Learning Conference (MSML'2022)*, PMLR **190**, pp. 1-16, Beijing, China, August, 2022.

219. A. RAFID, J. YIN, Y. GENG, S. LIANG, F. BAO, L. JU, AND G. ZHANG, A scalable training-free diffusion model for uncertainty quantification, *Workshop Proceedings of International Conference for High Performance Computing, Networking, Storage and Analysis (SC'2024)*, pp. 380-386, Atlanta, GA, November, 2024.
220. Z. ZENG, Y. WANG, L. JU, AND T. GUAN, Frequency-aware density control via reparameterization for high-quality rendering of 3D Gaussian splatting, *Proceedings of Thirty-Ninth AAAI Conference on Artificial Intelligence (AAAI'2025)*, 39(9), pp. 9833-9841, Philadelphia, PA, February-March, 2025.
221. C. RUAN, Y. WANG, T. GUAN, B. ZHANG, AND L. JU, IndoorGS: Geometric cues guided gaussian splatting for indoor scene reconstruction, *Proceedings of Thirty-Eighth IEEE Conference on Computer Vision and Pattern Recognition (CVPR'2025)*, pp. 844-853, Nashville, TN, June, 2025.
222. Z. ZENG, Y. WANG, C. YANG, T. GUAN, L. JU, Instant GaussianImage: A Generalizable and Self-Adaptive Image Representation via 2D Gaussian Splatting, *Proceedings of Twentieth International Conference on Computer Vision (ICCV'2025)*, pp. 27896-27905, Honolulu, HI, October, 2025.

Book chapters

223. L. JU, T. RINGLER AND M. GUNZBURGER, Voronoi diagrams and their application to climate and global modeling, in *Numerical Techniques for Global Atmospheric Models*, Lecture Notes in Computational Science and Engineering, Vol. **80**, pp. 313-342, Ed. by P. Lauritzen, C. Jablonowski, M. Taylor, R. Nair, Springer, 2011.

INVITED SEMINARS AND COLLOQUIA

1. Applied Mathematics Seminar, Department of Mathematics, University of North Carolina at Chapel Hill, 2002/1
2. Colloquium, Department of Mathematics, University of Pittsburgh, 2002/2
3. IMA/MCIM Industrial Problems Seminar, School of Mathematics, University of Minnesota, 2003/2
4. Iowa PDE Seminar, Department of Mathematics, Iowa State University, 2003/4
5. Applied and Computational Mathematics Seminar, Department of Mathematics, Pennsylvania State University, 2003/5
6. CSCAMM Seminar, Center for Scientific Computation and Mathematical Modeling, University of Maryland at College Park, 2003/10
7. CSIT Seminar Series, School of Computational Science & Information Technology, Florida State University, 2003/11
8. Colloquium, Department of Mathematics, University of South Carolina, 2004/1
9. Colloquium, Department of Mathematics and Statistics, Texas Tech University, 2004/2
10. Faculty Forum in Scientific Computing, University of South Carolina, 2005/2
11. PDEs and Numerical Methods Seminar, Department of Mathematics, Pennsylvania State University, 2005/3
12. Analysis and Computational Mathematics Seminar, Department of Mathematical Sciences, Clemson University, 2005/4
13. SIAM Student Chapter Seminar, Department of Mathematics, University of South Carolina, 2005/9
14. CNA Seminar, Department of Mathematical Sciences, Carnegie Mellon University, 2006/2

15. Seminar, Institute of Computational Mathematics & Scientific/Engineering Computing, Chinese Academy of Sciences, China, 2006/6
16. Colloquium, Research and Development Center for Parallel Software, Institute of Software, Chinese Academy of Sciences, China, 2006/6
17. SCS Seminar Series, School of Computational Science, Florida State University, 2006/11
18. Computational Mathematics Seminar, Department of Mathematical Sciences, Clemson University, 2007/4
19. BK21 Seminar, Department of Mathematics, Ajou University, Korea, 2007/5
20. Colloquium, School of Mathematics and System Sciences, Shandong University, China, 2007/5
21. Computational Nano Seminar, NanoCenter, University of South Carolina, 2007/11
22. Applied and Computational Mathematics Seminar, Department of Mathematical Sciences, George Mason University, 2008/3
23. Colloquium, Department of Mathematics, Xiangtan University, China, 2008/5
24. Colloquium, Department of Mathematics and Systems Science, National University of Defense Technology, China, 2008/5
25. Computational and Applied Mathematics Seminar, Department of Mathematics, Purdue University, 2008/10
26. Colloquium, Department of Mathematics and Statistics, Auburn University, 2008/12
27. SIAM Student Chapter Seminar, Department of Mathematics and Statistics, Auburn University, 2008/12
28. Seminar, Division of Mathematical Sciences, Nanyang Technological University, Singapore, 2008/12
29. Seminar, State Key Lab of Computer-Aided Design&Computer Graphics, Zhejiang University, China, 2009/7
30. Seminar, School of Mathematics and System Sciences, Shandong University, China, 2009/7
31. Aubrey Lecture Series, Department of Earth and Ocean Sciences, University of South Carolina, 2009/10
32. Applied Mathematics Seminar, Department of Mathematics, Ohio State University, 2010/4.
33. School of Sciences, Linyi Normal University, China, 2010/6
34. Seminar, School of Mathematical Sciences, Beijing Normal University, China, 2010/6
35. Department of Mathematics, Shanghai Jiaotong University, China, 2010/6
36. CCMA Luncheon Seminar, Department of Mathematics, Pennsylvania State University, 2010/11
37. Computational and Applied Mathematics Colloquium, Department of Mathematics, Pennsylvania State University, 2010/11
38. Applied and Computational Mathematics Seminar, School of Mathematics, Georgia Institute of Technology, 2011/4
39. Seminar, School of Mathematics, Shandong University, China, 2011/6
40. Seminar, Laboratory of Computational Geodynamics, Graduate University of Chinese Academy of Sciences, China 2011/6
41. Seminar, Center for Computational Science & Engineering, Peking University, China, 2011/6
42. Seminar, Institute of Computational Mathematics & Scientific/Engineering Computing, Chinese Academy of Sciences, China, 2011/6
43. Seminar, School of Mathematics, Shandong University, China, 2011/12

44. Seminar, Institute of Computational Mathematics & Scientific/Engineering Computing, Chinese Academy of Sciences, China, 2012/6
45. Seminar, School of Mathematics, Capital Normal University, China, 2012/6
46. Seminar, School of Mathematical Sciences, Beihang University, China, 2012/7
47. Seminar, School of Sciences, Shanghai Normal University, China, 2012/7
48. Applied Mathematics Seminar, Department of Mathematics, Michigan State University, 2012/9
49. Seminar, Institute of Computational Mathematics & Scientific/Engineering Computing, Chinese Academy of Sciences, China, 2013/5
50. Seminar, School of Mathematical Sciences, University of Science and Technology of China, China, 2013/5
51. Seminar, School of Mathematical Sciences, Beihang University, China, 2013/6
52. Seminar, Laboratory of Computational Geodynamics, Graduate University of Chinese Academy of Sciences, China, 2013/6
53. Colloquium, Department of Mathematics and Statistics, Missouri University of Science and Technology, 2013/11
54. ACM Seminar, Beijing International Center for Mathematical Research, Peking University, China, 2013/12
55. Seminar, Institute of Computational Mathematics & Scientific/Engineering Computing, Chinese Academy of Sciences, China, 2013/12
56. Seminar, School of Mathematical Sciences, Beihang University, China, 2014/6
57. Colloquium, Department of Mathematical Sciences, Shandong Normal University, China, 2014/5
58. Seminar, School of Mathematics, Shandong University, China, 2014/7
59. Seminar, State Key Lab of Computer-Aided Design&Computer Graphics, Zhejiang University, China, 2014/7
60. Seminar, School of Mathematical Sciences, Beijing Normal University, China, 2014/7
61. Seminar, Morningstar Mathematical Center, Chinese Academy of Sciences, China, 2014/7
62. Colloquium, Department of Mathematics, University of Alabama, 2014/9
63. Colloquium, Department of Mathematics and Statistics, Old Dominion University, 2014/10
64. CAM Seminar, Department of Mathematics, Iowa State University, 2014/10
65. Seminar, Chinese Network Information Center, Chinese Academy of Sciences, China, 2014/12
66. Seminar, School of Mathematics, Wuhan University, China, 2015/5
67. Seminar, School of Mathematics, Shanghai Normal University, China, 2015/6
68. Seminar, School of Mathematics, Sichuan University, China, 2015/7
69. Colloquium, Department of Applied Mathematics, Illinois Institute of Technology, 2015/9
70. Colloquium, Department of Applied Mathematics, Hong Kong Polytechnic University, 2015/10
71. Colloquium, Department of Mathematics, University of Tennessee at Chattanooga, 2015/11
72. Seminar, Institute of Mathematical Sciences, Renmin University of China, China, 2015/12
73. Seminar, Institute of Computational Mathematics & Scientific/Engineering Computing, Chinese Academy of Sciences, China, 2015/12
74. Applied Mathematics Seminar, Department of Mathematics, George Washington University, 2016/3

75. Seminar, State Key Laboratory of Cryospheric Sciences, Chinese Academy of Sciences, China, 2016/5
76. Colloquium, School of Mathematical Sciences, Lanzhou University, China, 2016/5
77. Seminar, Institute of Computational Mathematics & Scientific/Engineering Computing, Chinese Academy of Sciences, China, 2016/5
78. Colloquium, Department of Applied Mathematics, Hong Kong Polytechnic University, Hong Kong, 2016/6
79. Seminar, Department of Mathematical Sciences, Tsinghua University, China, 2016/6
80. Seminar, School of Mathematics, Shandong University, China, 2016/7
81. Seminar, School of Mathematical Sciences, Beihang University, China, 2016/7
82. Seminar, School of Mathematics, Wuhan University, China, 2016/7
83. Seminar, School of Applied Mathematics and Sciences, Beijing University of Technology, China, 2016/7
84. Seminar, Department of mathematical sciences, Clemson University, 2016/9
85. Colloquium, Department of Computational Mathematics, Science and Engineering, Michigan State University, 2016/10
86. Seminar, Institute of Computational Mathematics & Scientific/Engineering Computing, Chinese Academy of Sciences, China, 2016/12
87. Seminar, School of Mathematics, Wuhan University, China, 2016/12
88. Colloquium, Oak Ridge National Laboratory, 2017/2.
89. CAM Seminar, Department of Mathematics, University of Tennessee at Knoxville, 2017/2.
90. Appl. Math. Seminar, Department of Mathematics, University of Wisconsin at Madison, 2017/4
91. Colloquium, School of Science, Wuhan University of Science and Technology, China, 2017/5
92. Seminar, School of Mathematics, Wuhan University, China, 2017/5
93. Colloquium, School of Mathematical Sciences, Electric University of Science and Technology of China, 2017/5
94. Seminar, Center for Mathematical Sciences, Huazhong University of Science and Technology, China, 2017/7
95. Seminar, School of Mathematical Sciences, Shanghai Jiao Tong University, China, 2017/7
96. Seminar, Department of Mathematical Sciences, Michigan Technological University, 2017/9
97. Seminar, School of Mathematical Sciences, Jilin University, China, 2017/9
98. Seminar, School of Mathematics, Shandong University, China, 2017/10
99. Colloquium, School of Mathematics, Shandong Normal University, China, 2017/10
100. Seminar, School of Mathematical Sciences, Beihang University, China, 2017/10
101. Numerical Analysis Seminar, Department of Mathematics, University of Hong Kong, Hong Kong, 2017/12
102. Seminar, School of Mathematics and Statistics, Central China Normal University, China, 2017/12
103. Seminar, Institute of Computational Mathematics & Scientific/Engineering Computing, Chinese Academy of Sciences, China, 2018/1
104. Seminar, China Network Information Center, Chinese Academy of Sciences, China, 2018/1
105. Seminar, School of Sciences, Shanghai Normal University, China, 2018/1

106. Seminar, School of Mathematical Sciences, Tongji University, China, 2018/1
107. Seminar, School of Mathematical Sciences, Shanghai Jiao Tong University, China, 2018/4
108. Colloquium, School of Mathematical Sciences, Beihang University, China, 2018/4
109. Seminar, School of Sciences, Nanchang University, China, 2018/4
110. Seminar, School of Mathematical Sciences, Ocean University of China, China, 2018/4
111. Seminar, School of Mathematical Sciences, Beijing Normal University, China, 2018/5
112. Seminar, China Network Information Center, Chinese Academy of Sciences, China, 2018/5
113. Seminar, Department of Mathematics, Southern University of Science and Technology of China, 2018/6
114. Seminar, School of Data and Computer Science, Sun Yat-sen University, China, 2018/6
115. Seminar, School of Applied Mathematics and Sciences, Beijing University of Technology, China, 2018/7
116. Applied Mathematics Seminar, Department of Mathematics, Ohio State University, 2018/10
117. Seminar, School of Mathematics, Shandong University, China, 2018/10
118. Colloquium, School of Mathematics and Statistics, Huazhong University of Science and Technology, China, 2019/1
119. CAM Seminar, Department of Mathematics and Statistics, Mississippi State University, 2019/4
120. Seminar, Yau Mathematical Sciences Center, Tsinghua University, China, 2019/5
121. Colloquium, School of Mathematics and Statistics, Zhengzhou University, China, 2019/5
122. Seminar, School of Mathematics and Econometrics, Hunan University, China, 2019/5
123. Seminar, School of Mathematics, Shandong University, China, 2019/5
124. Seminar, School of Mathematical Sciences, Ocean University of China, China, 2019/5
125. Seminar, School of Sciences, Shanghai Normal University, China, 2019/7
126. Seminar, School of Mathematical Sciences, East China Normal University, China, 2019/7
127. Seminar, School of Mathematical Sciences, Shanghai Jiao Tong University, China, 2019/7
128. Colloquium, Department of Mathematics and Statistics, Texas Tech University, 2019/10
129. Seminar, Department of Mathematics, Ajou University, Korea, 2019/11
130. Seminar, School of Mathematical Sciences, Beihang University, China, 2019/12
131. Colloquium, School of Computer Science and Engineering, South China University of Technology, China, 2019/12
132. Seminar, Research Center for Applied Mathematics and Interdisciplinary Studies, South China Normal University, China, 2019/12
133. Seminar, Department of Applied Mathematical, Hong Kong Polytechnic University, Hong Kong, 2020/5 (Online talk)
134. Finite Element Seminar Series, Tianyuan Mathematical Center in Northeast China, Jilin University, China, 2020/6 (Online talk)
135. Computational Mathematics Seminar, School of Mathematical Sciences, East Normal University of China, China, 2020/7 (Online talk)
136. Colloquium, School of Applied Mathematics and Sciences, Beijing University of Technology, China, 2020/7 (Online talk)
137. Zhufeng Colloquium, School of Mathematics, Shandong University, China, July, 2020/10 (Online talk)

138. Zhufeng Colloquium, School of Mathematics, Shandong University, China, June, 2021/6 (Online talk)
139. Seminar, School of Mathematics and Statistics, Wuhan University, China, July, 2021/7 (Online talk)
140. Applied and Computational Mathematics Seminar, School of Mathematics, Georgia Institute of Technology, 2021/10 (Online talk)
141. Colloquium, Department of Applied Mathematical, Hong Kong Polytechnic University, Hong Kong, 2021/11 (Online talk)
142. Oversea Scholars Forum, School of Mathematics and Statistics, Henan University, China, 2021/11 (Online talk)
143. Colloquium, School of Mathematics, Shandong University, China, 2021/12 (Online talk)
144. Seminar, School of Mathematics, Hunan University, China, 2021/12 (Online talk)
145. Seminar, School of Mathematics and Statistics, Zhengzhou University, China, 2021/12 (Online talk)
146. Colloquium, School of Mathematics and Statistics, Shandong Normal University, China, 2022/1 (Online talk)
147. Colloquium, Beijing Computational Science Research Center, China, 2022/2 (Online talk)
148. Seminar, Institute of Computational Mathematics & Scientific/Engineering Computing, Chinese Academy of Sciences, China, 2022/5 (Online talk)
149. Seminar, School of Mathematics and Statistics, Wuhan University, China, 2022/5 (Online talk)
150. Colloquium, Department of Applied Mathematics, Hong Kong Polytechnic University, Hong Kong, 2022/6 (Online talk)
151. Numerical Analysis Seminar, Department of Mathematics, University of Hong Kong, Hong Kong, 2022/9 (Online talk)
152. Seminar, School of Mathematical Sciences, Tongji University, 2022/10 (Online talk)
153. Seminar, School of Mathematical Sciences, Ocean University of China, China, 2022/11 (Online talk)
154. Colloquium, Department of Mathematics and Statistics, Missouri University of Science and Technology, 2023/2
155. Colloquium, School of Mathematical Sciences, Shanghai Jiao Tong University, China, 2023/3 (Online talk)
156. Zhufeng Colloquium, School of Mathematics, Shandong University, China, 2023/6
157. Seminar, School of Mathematics and Statistics, Wuhan University, China, 2023/6
158. Seminar, School of Applied Mathematics and Sciences, Beijing University of Technology, China, 2023/6
159. ACM Seminar, Beijing International Center for Mathematical Research, Peking University, China, 2023/6
160. Seminar, Institute of Computational Mathematics & Scientific/Engineering Computing, Chinese Academy of Sciences, China, 2023/6
161. Seminar, College of Mathematics and Science, Shanghai Normal University, China, 2023/6
162. CAM Colloquium, Department of Mathematics, Pennsylvania State University, 2023/10
163. Colloquium, Tianyuan Mathematical Center in Central China, Wuhan University, China, 2023/12
164. Applied and Computational Mathematics Seminar, School of Mathematics, Georgia Institute of Technology, 2024/2

165. Applied Mathematics Seminar, Department of Mathematics, University of Georgia, 2024/4
166. Seminar, School of Mathematics, Yunnan Normal University, China, 2024/5
167. Seminar, School of Statistics, Mathematics, Yunnan University of Finance and Economics, China, 2024/5
168. Colloquium, School of Mathematics, Peking University, China, 2024/5
169. Seminar, School of Mathematical Sciences, Beijing Normal University, China, 2024/5
170. Seminar, Institute of Computational Mathematics & Scientific/Engineering Computing, Chinese Academy of Sciences, China, 2024/5
171. Seminar, School of Applied Mathematics and Sciences, Beijing University of Technology, China, 2023/6
172. Seminar, School of Mathematics and Systems Science, Xinjiang University, China, 2024/6
173. Seminar, School of Mathematical Sciences, Ocean University of China, China, 2024/6
174. Numerical Analysis Seminar, Department of Mathematics, University of Hong Kong, Hong Kong, 2024/7
175. Seminar, School of Mathematics and Statistics, Huazhong University of Science and Technology, China, 2024/8
176. Seminar, School of Mathematics and Statistics, Central China Normal University, China, 2024/8
177. Seminar, School of Mathematics and Computer Science, Nanchang University, China, 2024/8
178. SSE Seminar, Department of Mathematics, Chinese University of Hong Kong at Shenzhen, China, 2024/9
179. Seminar, School of Mathematics, Tongji University, China, 2024/9
180. Zhufeng Colloquium, School of Mathematics, Shandong University, China, 2024/9
181. Seminar, School of Mathematics and Statistics, Shandong Normal University, China, 2024/9
182. Applied and Computational Mathematics Seminar, Department of Mathematics and Statistics, Auburn University, 2024/10
183. Colloquium, School of Mathematics and Statistics, Zhengzhou University, China, 2024/11
184. Seminar, School of Mathematics and Statistics, Henan University, China, 2024/11
185. Colloquium, School of Mathematics and Statistics, Hunan Normal University, China, 2024/11
186. Seminar, School of Mathematical Sciences, Shanghai Jiao Tong University, China, 2024/12
187. Colloquium, Department of Mathematics, University of Alabama, 2025/3
188. CCAM Seminar, Department of Mathematics, Purdue University, 2025/9
189. Colloquium, Department of Mathematics, Michigan State University, 2025/10
190. Computational Mathematics Seminar, School of Mathematical and Statistical Sciences, Clemson University, 2026/2
191. Applied Math Seminar, Department of Mathematics, University of Buffalo, 2026/4
192. Seminar, Institute of Computational Mathematics & Scientific/Engineering Computing, Chinese Academy of Sciences, China, 2026/5
193. Seminar, Institute of Computational Mathematics & Scientific/Engineering Computing, Chinese Academy of Sciences, China, 2026/5
194. Seminar, School of Mathematical Sciences, Beijing Normal University, China, 2026/6
195. Seminar, Academy for Multidisciplinary Studies, Capital Normal University, China, 2026/6

CONFERENCE/WORKSHOP PRESENTATIONS

1. Emerging Methodologies and Applications in Numerical PDEs, Tallahassee, FL, March 2004
2. The Second IEEE International Symposium on Biomedical Imaging, Arlington, VA, April 2004
3. SIAM Annual Meeting 2005, New Orleans, LA, July 2005
4. SIAM Southeastern Atlantic Section Annual Meeting 2006, Auburn University, Auburn, AL, March 2006
5. International Conference on Recent Advance in Scientific Computation, Beijing, China, June 2006
6. SIAM Conference on Computational Science and Engineering 2007, Costa Mesa, CA, February 2007
7. SIAM Annual Meeting 2008, San Diego, CA, July 2008
8. SIAM Conference on Computational Science and Engineering 2009, Miami, FL, March, 2009
9. SIAM Southeastern Atlantic Section Annual Meeting 2009, University of South Carolina, Columbia, SC, April 2009
10. Spring AMS Southeastern Section Meeting 2010, University of Kentucky, Lexington, KY, March 2010
11. SIAM Southeastern Atlantic Section Annual Meeting 2010, North Carolina State University, Raleigh, NC, March 2010
12. Scientific Computing and Nonlinear Partial Differential Equations, Jiuzhaigou, Sichuan Province, China, June 2010
13. SIAM Southeastern Atlantic Section Annual Meeting 2011, University of North Carolina, Charlotte, NC, March 2011
14. 2011 Spring AMS Western Section Meeting, University of Nevada, Las Vegas, NV, April 2011
15. International Conference on Applied Mathematics and Interdisciplinary Research, Tianjin, China, June 2011
16. The Seventh International Congress on Industrial and Applied Mathematics, Vancouver, Canada, July 2011
17. SIAM Southeastern Atlantic Section Annual Meeting 2012, University of Alabama Huntsville, Huntsville, AL, March 2012
18. International Conference on Scientific Computing and Applications 2012, University of Nevada, Las Vegas, NV, April 2012
19. The Second International Conference on Scientific Computing, Nanjing Normal University, Nanjing, China, May 2012
20. International Workshop on Computational and Applied Mathematics, CSRC, Beijing, China, June 2012
21. International Workshop on Numerical Methods for Partial Differential Equations, Shandong Normal University, Jinan, China, June 2012
22. Nonlocal Continuum Models for Diffusion, Mechanics, and Other Applications, SAMSI, Durham, NC, June 2012
23. International Conference on Mathematical Modeling, Analysis and Computation, Xiamen University, Xiamen, China, July 2012
24. AMS-MAA Joint Mathematics Meetings 2013, San Diego, CA, January 2013
25. CESM-LIWG Winter Meeting 2013, Boulder, CO, February 2013

26. SIAM Conference on Computational Science and Engineering 2013, Boston, MA, February-March 2013
27. SIAM Southeastern Atlantic Section Annual Meeting 2013, Knoxville-ORNL, TN, March 2013
28. International Conference on Mathematical Modeling and Computation, Wuhan University, Wuhan, China, May 2013
29. International Workshop on Computational and Applied Mathematics, Yellow Mountain, China, May 2013
30. The Second International Conference on Engineering and Computational Mathematics, Hong Kong Polytechnic University, Hong Kong, December 2013
31. Spring AMS Southeastern Section Meeting 2014, University of Tennessee, Knoxville, TN, March 2014
32. SIAM Southeastern Atlantic Section Annual Meeting 2014, Florida Institute of Technology, Melbourne, FL, March 2014
33. The 2014 International Conference on Modeling of Complex Biological Systems, Tianjin, China, May 2014
34. International Workshop on the Frontiers of Computational Geodynamics, Beijing, China, July 2014
35. Fall AMS Southeastern Section Meeting 2014, University of North Carolina Greensboro, Greensboro, NC, November, 2014
36. SIAM Conference on Computational Science and Engineering 2015, Salt Lake City, Utah, March 2015
37. SIAM Southeastern Atlantic Section Annual Meeting, University of Alabama, Birmingham, AL, March 2015
38. SIAM Central States Section Annual Meeting 2015, Missouri University of Science and Technology, Rolla, MO, April 2015
39. International Conference on Numerical Partial Differential Equations and their Applications, Wuhan University, Wuhan, China, May 2015
40. The Eighth International Congress on Industrial and Applied Mathematics 2015, Beijing, China, August 2015
41. Workshop on Computational Mathematics and Scientific Computing, Jeju Island, South Korea, August 2015
42. Workshop on Advances in Scientific Computing and Applied Mathematics, University of Nevada, Las Vegas, NV, October 2015
43. SIAM Southeastern Atlantic Section Annual Meeting 2016, University of Georgia, Athens, GA, March 2016
44. International Conference on Applied Mathematics, City University of Hong Kong, Hong Kong, May-June 2016
45. Workshop on Computational Mathematics and Application Advances, Jiujiang, China, July 2016
46. SIAM Central States Section Annual Meeting 2016, University of Arkansas, Little Rock, AK, September-October 2016
47. IMACS 2016 World Congress, Xiamen, China, December 2016
48. Workshop on Current Trends in Numerical PDEs and Applications, Northwestern Polytechnical University, Xi'an, China, December 2016
49. AMS-MAA Joint Mathematics meeting 2017, Atlanta, January 2017

50. SIAM Conference on Computational Science and Engineering 2017, Atlanta, GA, March 2017
51. SIAM Southeastern Atlantic Section Annual Meeting 2017, Florida State University, Tallahassee, FL, March 2017
52. Workshop on Advances of Scientific Computing, Nanchang University, Nanchang, China, May 2017
53. Third International Conference on Engineering and Computational Mathematics, Hong Kong, May-June 2017
54. Workshop on Phase Field Method and Applications, Beijing International Center of Mathematical Research, China, July 2017
55. The 16th International Workshop on Multi-scale (Un)-structured Mesh Numerical Modeling, Stanford University, CA, August 2017
56. Workshop on Complex Systems and High-Performance Computing, Xuchang University, Xuchang, China, September 2017
57. Workshop on Computational Mathematics, Hong Kong Polytechnic University, Hong Kong, December 2017
58. International Conference on Recent Advances in Computational and Applied Mathematics, Wuhan University, Wuhan, China, December 2017
59. Workshop on Modeling and Algorithms for Ocean Simulations, LSEC, Beijing, China, May 2018
60. Workshop on High Accuracy Schemes and Conservative Schemes for PDEs, Qingdao, China, June 2018
61. The Twelfth AIMS Conference on Dynamical Systems, Differential Equations and Applications, Taipei, Taiwan, July 2018
62. Workshop on High Performance and Scientific Computing, Qingdao, China, October 2018
63. Workshop on Mathematical and Computational Methods for Quantum Systems, University of Montreal, Montreal, Canada, December 2018
64. International Youth Forum on Computational Mathematics, Xiangtan University, Xiangtan, China, December 2018
65. SIAM Conference on Computational Science and Engineering 2019, Spokane, WA, February-March 2019
66. Workshop on Numerical Computing and Applications, Hunan Normal University, Changsha, China, May 2019
67. International Conference on Advanced Numerical Methods for Scientific Computation, Southern University of Science and Technology, Shenzhen, China, June 2019
68. International Workshop on Numerical Methods, Scientific Computing and Applications, Yantai University, Yantai, China, July 2019
69. The Twelfth China National Conference on Computational Mathematics, Harbin, China, July-August 2019 (**Plenary speaker**)
70. SIAM Southeastern Atlantic Section Annual Meeting 2019, University of Tennessee, Knoxville, TN, September 2019
71. The ICCV Workshop on Moving Cameras, Seoul, Korea, October 2019
72. AMS-MAA Joint Mathematics Meeting 2020, Denver, CO, January 2020
73. Scientific Computing Around Louisiana (SCALA) 2020, Louisiana State University, Baton Rouge, LA, February 2020 (**Keynote speaker**)

74. Workshop on Recent Progress in Nonlocal Modeling, Analysis and Computation, June 2020 (Visual conference)
75. OUC Workshop on Computational Mathematics, July 2020 (Visual conference)
76. DOE ESMD-E3SM PI Meeting, October 2020 (Visual conference)
77. Workshop on Recent Progress in Phase Field Related Problems, November 2020 (Visual conference)
78. SIAM Conference on Computational Science and Engineering 2021, March 2021 (Visual conference)
79. Nonlocal Model: Theory and Computation, Wuhan University, Wuhan, China, April 2021 (Hybrid conference)
80. SIAM Southeastern Atlantic Section Annual Meeting 2021, Auburn University, Auburn, AL, September 2021 (Hybrid conference)
81. International Workshop on Physical Property-Preserving Numerical Methods for PDEs, Ocean University of China, Qingdao, China, November 2021 (Visual conference)
82. International Workshop on Computation, Analysis and Applications of PDEs with Nonlocal and Singular Operators, National University of Singapore, Singapore, February 2022 (Hybrid conference)
83. AMS-MAA Joint Mathematics Meeting 2022, April 2022 (Visual conference)
84. SIAM Conference on Uncertainty Quantification, Atlanta, GA, April 2022 (Hybrid conference)
85. International Conference on Computational Methods and Applications in Engineering, Mississippi State University, Starkville, MS, May 2022 (Hybrid conference)
86. The Twenty-Seventh CESM Workshop, June 2022 (Visual conference)
87. Copper Country Workshop on Applied Mathematics, Statistics, and Data Sciences, Michigan Technological University, Houghton, MI, July 2022
88. Workshop on Backward-Forward Stochastic Computing, China, August 2022 (Visual conference)
89. The Second SIAM Conference on Mathematics of Data Science, San Diego, CA, September 2022 (Hybrid conference)
90. Women Conference in Scientific Computing on Complex Physical and Biological Systems, University of Florida, Gainesville, October 2022
91. SIAM Southeastern Atlantic Section Annual Meeting 2023, Virginia Tech, Blacksburg, VA, March 2023 (**Plenary speaker**)
92. ASCR SciML Project Meeting, Columbia University, NY, April 2023
93. Workshop on Computational Mathematics Trends in Science and Engineering, University of Pittsburgh, Pittsburgh, PA, April 2023
94. The Seventh international Conference on Scientific Computing and Partial Differential Equations, Hong Kong Polytechnic University, Hong Kong, May 2023
95. The Tenth International Congress on Industrial and Applied Mathematics, Tokyo, Japan, August 2023
96. 2023 Fall AMS Eastern Sectional Meeting, University of Buffalo, Buffalo, NY, September 2023
97. The Forty-First Southeastern-Atlantic Regional Conference on Differential Equations, Florida A&M University, Tallahassee, FL, November 2023
98. Workshop on Recent Advances in Scientific Computing and Deep Learning, Tsinghua Sanya International Mathematics Forum, Sanya, Hainan, China, December 2023

99. 2024 Spring AMS Southeastern Sectional Meeting, Florida State University, Tallahassee, FL, March 2024
100. The Second International Conference on Mathematical Modeling and Numerical Methods, Peking University, Beijing, China, June 2024
101. International Workshop on Deep Learning and Numerical Methods for PDEs, Xi'an Jiao Tong University, Xi'an, China, June 2024
102. NSF Computational Mathematics PI Meeting 2024, University of Washington, Seattle, WA, July 2024
103. International Workshop on Structure Preserving Numerical Methods and their Applications, Hong Kong Polytechnic University, Hong Kong, July 2024
104. BNU Workshop on Computational Mathematics, Hunchun, Jilin, China, August 2024
105. SIAM Conference on Mathematics of Data Science 2024, Atlanta, GA, October 2024
106. Workshop on Novel Numerical Methods in Scientific Computing, Shandong University, Jinan, China, November 2024
107. Workshop on Computational Mathematics, Anhui University, Hefei, China, November 2024
108. Workshop on Mathematical Theory and Algorithms for Multiphysics Field Simulations, Beijing Normal University, Beijing, China, March 2025
109. SIAM Conference on Computational Science and Engineering 2025, Fort Worth, Texas, March 2025
110. Workshop on Recent Advances in Numerical PDEs, University of Pittsburgh, Pittsburgh, PA, May 2025
111. NSF Computational Math Program Meeting 2025, University of Utah, May 2025
112. VIASM Workshop on Advanced Numerical Methods for Deterministic and Stochastic Differential Equations, Hanoi, Vietnam, June 2025
113. The Fifteenth International Congress on Spectral and High Order Methods, Montreal, QC, Canada, July 2025
114. International Workshop on Advances in Scientific Computing and Applied Mathematics, Breckenridge, CO, October 2025
115. AMS-MAA Joint Mathematics Meetings 2026, Washington, D.C., January 2026
116. Spring AMS Southeastern Section Meeting 2026, Georgia Southern University-Armstrong Campus, Savannah, GA, March 2026
117. Workshop on Recent Advances in Numerical PDEs, University of Pittsburgh, Pittsburgh, PA, May 2026
118. SIAM Conference on Mathematics of Data Science 2026, Salt Lake City, Utah, November 2026
119. SIAM Conference on Computational Science and Engineering 2027, Salt Lake City, Utah, February 2027

POSTERS

1. Human Brain Mapping 2003 (HBM'03), New York, June 2003
2. Climate Change Prediction Program Meeting 2007, Indianapolis, IN, September 2007
3. Climate Change Prediction Program Meeting 2009, Bethesda, MD, April 2009
4. DOE Climate Change Modeling Program Science Team Meeting 2010, Gaithersburg, MD, March-April 2010
5. DOE ACME All-Hands Meeting, Potomac, MD, June 2017
6. DOE Climate Modeling PI's Meeting, Potomac, MD, November 2018

7. The Seventeenth International Conference on Computer Vision, Seoul, Korea, October-November 2019.
8. DOE ASCR Applied Mathematics PI Meeting 2024, Albuquerque, NM, January 2024
9. The First ORNL AI4Science Workshop, Oak Ridge National Lab, Oak Ridge, TN, April 2025
10. DOE ASCR Applied Mathematics PI Meeting 2026, Washington, D.C., July 2026

ORGANIZED MEETINGS

CONFERENCES/WORKSHOPS

1. Member of organizing committee, SIAM Southeastern Atlantic Section Annual Meeting 2008, Orlando, FL, March 2008
2. Co-Chair of organizing committee, SIAM Southeastern Atlantic Section Annual Meeting 2009, Columbia, SC, April 2009
3. Co-Chair of organizing committee, International Workshop on Computational and Applied Mathematics, Beijing, China, June 2012
4. Member of organizing committee, International Workshop on Complex Systems and High Performance Computing, Xuchang, China, September 2017
5. Co-Chair of organizing committee, Ninth Annual Graduate Student Mini-conference in Computational Mathematics, Columbia, SC, February 2018
6. Co-Chair of organizing committee, International Conference on Mathematical Modeling and Numerical Methods, Qingdao, China, May-June 2019
7. Member of organizing committee, International Conference on Mathematical Modeling and Numerical Methods, Beijing, China, June 2024

MINISYMPOSIA

1. Organizer, Minisymposium – “*Algorithms and Applications of Tessellations of Spaces and Surface*” (1 session), SIAM Southeastern Atlantic Section Annual Meeting 2006, Auburn, AL, March 2006
2. Co-Organizer, Minisymposium – “*Centroidal Voronoi Tessellations: Theory, Algorithms and Applications*” (2 sessions), SIAM Conference on Computational Science & Engineering 2007, Costa Mesa, CA, February 2007
3. Organizer, Minisymposium – “*Numerical Methods for Partial Differential Equations*” (2 sessions), SIAM Southeastern Atlantic Section Annual Meeting 2008, Orlando, FL, March 2008
4. Co-Organizer, Minisymposium – “*Recent Advances in Algorithms and Applications of Centroidal Voronoi Tessellation and Optimal Quantization*” (2 sessions), SIAM Annual Meeting 2008, San Diego, CA, July 2008
5. Co-Organizer, Minisymposium – “*Recent Advances in Studies and Applications of Centroidal Voronoi Tessellations*” (2 sessions), International Congress on Industrial and Applied Mathematics 2011, Vancouver, Canada, July 2011.
6. Co-Organizer, Minisymposium – “*Recent Advances in Numerical PDEs and Computational Biology*” (2 sessions), SIAM Southeastern Atlantic Section Annual Meeting 2012, Huntsville, AL, March 2012
7. Co-Organizer, Minisymposium – “*Recent Advances in Modeling of Complex Systems: Analysis and Computation*” (3 sessions), SIAM Southeastern Atlantic Section Annual Meeting 2014, Melbourne, FL, March 2014
8. Co-Organizer, Minisymposium – “*Modeling and simulations of Complex Biological Systems*” (4 sessions), International Congress on Industrial and Applied Mathematics 2015, Beijing, China, August 2015

9. Co-Organizer, Minisymposium – “*Recent Advances on Modeling, Analysis and Computation of Nonlocal Problems* ” (3 sessions), SIAM Southeastern Atlantic Section Annual Meeting 2017, Tallahassee, March 2017
10. Co-Organizer, Workshop – “*Numerical Methods for Phase Field Models*”, The Third International Conference on Engineering and Computational Mathematics, Hong Kong, May-June 2017
11. Organizer, Minisymposium – “*Numerical Partial Differential Equations and Applications*” (1 session), SIAM Southeastern Atlantic Section Annual Meeting 2018, Chapel Hill, NC, March 2018
12. Co-Organizer, Minisymposium – “*Classic and Deep Learning Methods for Data Driven Models*” (2 sessions), SIAM Southeastern Atlantic Section Annual Meeting 2021, Auburn, AL, September 2021
13. Co-Organizer, Minisymposium – “*Machine Learning for Large-Scale Scientific Data Analytics*” (2 sessions), SIAM Conference on Mathematics of Data Science 2022, San Diego, CA, September 2022
14. Co-Organizer, Minisymposium – “*Recent Advances in Numerical Methods for Multiscale and Multiphysics Partial Differential Equations*” (2 sessions), SIAM Southeastern Atlantic Section Annual Meeting 2023, Blacksburg, VA, March 2023

NSF PANELISTS

- NSF-DMS Panelist for Comput. Math. Program, 2008, 2010, 2013, 2014, 2016, 2023, 2025
- NSF-DMS Panelist for Career Program, 2012

REFeree FOR PROPOSALS

- National Science Foundation
- Department of Energy
- Singapore National Research Foundation
- Ministry of Education Research Grant, Singapore
- Hong Kong Research Grants Council
- National Research Foundation, United Arab Emirates
- KAUST Competitive Research Grants

RESEARCH SUPERVISION

GRADUATE STUDENTS

• DOCTORAL STUDENTS

- ☐ Li Tian (USC/Math), Ph.D., 2009/8
Dissertation: “*Error Estimates for Finite Element/Volume Approximations of Dissipative Partial Differential Equations on Surfaces*”
- ☐ Xiao Xiao (USC/Math), Ph.D., 2012/8
Dissertation: “*Covolume-upwind Finite Volume Approximations for Linear Elliptic Partial Differential Equations*”
- ☐ Yu Cao (USC/CSE), Ph.D. (co-supervisor), 2013/8
Dissertation: “*3D Grain Segmentation in Superalloy Images Using Multichannel Edge-Weighted Centroidal Tessellation Based Methods*”
- ☐ Youjie Zhou (USC/CSE), Ph.D. (co-supervisor), 2015/8

Dissertation: *“Propagated Image Segmentation Using Edge-Weighted Centroidal Voronoi Tessellation Based Methods”*

- Chenfei Zhang (USC/Math), Ph.D., 2019/8

Dissertation: *“Unconditionally Energy Stable Linear Schemes for a Two-Phase Diffuse Interface Model with Peng-Robinson Equation of State”*

- Shuai Yuan (USC/Math), Ph.D. (co-supervisor), 2020/8

Dissertation: *“An Ensemble-based Projection Method and its Numerical Investigation”*

- Zhenyao Wu (USC/CSE), Ph.D. (co-supervisor), 2022/8

Dissertation: *“Learning Depth from Images”*

- Xinyi Wu (USC/CSE), Ph.D. (co-supervisor), 2022/8

Dissertation: *“Cross Domain Semantic Segmentation”*

- Yuankai Teng (USC/Math), Ph.D., 2023/8

Dissertation: *“Deep Learning Methods for Some Problems in Scientific Computing”*

- Yuwei Geng (USC/Math), Ph.D. (co-supervisor), 2026/8

Dissertation: *“Operator Learning for Energy Conservative and Energy Dissipative Systems”*

- Anusooya Vasudevan (USC/Math), Ph.D., expected 2027/8

- Sabrina Rashid (USC/Math), Ph.D. (co-supervisor), expected 2027/8

- Shuting Wang (USC/Math), Ph.D., current.

• MASTERS STUDENTS

- Wensong Wu (USC/Math), M.S., 2007/8

Thesis: *“Adaptive Finite Volume Methods for Convection-Diffusion Equations with Mesh Optimization”*

- Xiao Xiao (USC/Math), M.S., 2010/5

Thesis: *“Over-relaxation Lloyd Method for Computing Centroidal Voronoi Tessellations”*

- Jing Liu (USC/Math), M.S., 2010/8

Thesis: *“Construction of Centroidal Voronoi Tessellations Using a Conjugate Gradient Method Based on Trust Regions”*

- Rosalia Tatano (USC/Math), M.S., 2013/5

Thesis: *“Study on Covolume-Upwind Finite Volume Approximations for Linear Parabolic Partial Differential Equations”*

- Meshack Kiplagat (USC/Math), M.S., 2013/12

Thesis: *“The Compact Implicit Integration Factor Scheme for the Solution of Allen-Cahn Equation”*

POSTDOCS

- Huai Zhang, Postdoc, 2008/9-2010/3
- Xiaoran Shi, Postdoc, 2012/7-2013/7
- Hao Tian, Postdoc, 2013/7-2016/1
- Tong Zhang, Postdoc, 2013/8-2015/8
- Jianfang Lu, Postdoc, 2016/7-2018/8
- Thi-Thao-Phuong Hoang, Postdoc, 2017/1-2018/8
- Huadong Gao, Postdoc, 2018/3-2018/12

- Xiao Li, Postdoc, 2018/8-2019/8
- Xucheng Meng, Postdoc, 2018/9-2019/8
- Anthony Gruber, Postdoc, 2021/1-2022/8
- Rihui Lan, Postdoc, 2020/5-2022/11

VISITING SCHOLARS

- Yang Li, Visiting Student, 2010/8-2011/2
- Weidong Zhao, Visiting Scholar, 2013/9-2013/11
- Wei Leng, Visiting Scholar, 2013/9-2013/12
- Liyong Zhu, Visiting Scholar, 2014/3-2015/3
- Huai Zhang, Visiting Scholar, 2015/8-2015/9
- Jian Zhang, Visiting Scholar, 2015/10-2015/11
- Hongwei Li, Visiting Scholar, 2016/9-2017/9
- Chunbao Zhou, Visiting Scholar, 2017/1-2017/8
- Xiaoping Zhang, Visiting Scholar, 2018/6-2018/8
- Jianguo Huang, Visiting Scholar, 2018/8-2018/9
- Zhaocan Ma, Visiting Student, 2019/2-2019/5
- Kun Jiang, Visiting Student, 2019/9-2020/9
- Jingwei Li, Visiting Student, 2019/10-2020/10
- Jiayi Duan, Visiting Student, 2025/1-2025/7

TEACHING AT USC

NUMBER	COURSE TITLE	CRS.	Terms
MATH 141	Calculus I	4	S06, S10, F11
MATH 142	Calculus II	4	S12, S14, F19, F20, F21, F22, F23
MATH 241	Vector Calculus	3	F04, F12, F14, F18
MATH 242	Elementary Differential Equations	3	S09, S13, S15, S16, S19, S20, S22, S24
MATH 344	Applied Linear Algebra	3	F18, S19, S24, F25, F26
MATH 514	Financial Mathematics I	3	F13, F19, F21, F23, F25, F26
MATH 515	Financial Mathematics II	3	S14
MATH 520	Ordinary Differential Equations	3	F07, F15
MATH 526	Numerical Linear Algebra	3	S05, S06, F06, F08, F09, S12, S16
MATH 527	Numerical Analysis	3	S10
MATH 550	Vector Analysis	3	S20, S21, S26
MATH 554	Analysis I	3	F22
MATH 708	Foundations of Comput. Math. I	3	F11, F12
MATH 709	Foundations of Comput. Math. II	3	S13, S20
MATH 723	Differential Equations I	3	F14, F16
MATH 724	Differential Equations II	3	S15, S17
MATH 726	Numerical Analysis I	3	F06
MATH 727	Numerical Analysis II	3	S07
MATH 727	Numerical Differential Equations II	3	S23, S25
MATH 728J	Scientific Computing I	3	F05

SERVICE

DEPARTMENT

- Chair of Committee of the Full Professors (2021/4 – 2022/4)
- Chair of Committee of the Tenured Faculty (2009/4 – 2010/4)
- Co-Chair of the Bylaws Committee (2023/4 – 2024/4)
- Post-Tenure Review Committee (2008/8 – 2009/5, 2011/8 – 2013/5, 2014/8 – 2016/5, 2022/8 – 2024/8 (Chair))
- Undergraduate Advisory Council (2004/8 – 2006/8)
- Graduate Advisory Council (2006/8 – 2007/8, 2009/8 – 2010/8, 2011/8 – 2019/8)
- Faculty Advisory Council (2012/8 – 2014/8, 2017/8 – 2022/8, 2025/8–Present)
- Hiring Committee (2005/8 – 2006/8, 2007/8 – 2009/8, 2023/8 – 2024/8)
- BioMath Hiring Committee (2012/8 – 2013/8)
- Applied and Computational Mathematics Committee (2004/8 – 2010/8, 2011/8 – Present)
- Calculus II Coordinator (2020/8 – 2024/8)
- Graduate Placement Committee (2011/8 – 2013/8)
- Peer Review of Teaching Committee (2016/9 – 2023/8, 2025/1 – Present)
- Grant Mentoring Committee (2009/8 – 2010/8)
- Computer Committee (2004/8 – 2010/8)
- Service Courses Committee (2025/1–2025/8)
- Strategic Hiring Plan Committee (2026/1–Present)
- Faculty Advisor of SIAM Student Chapter at USC (2006/8 – 2008/8)
- Organizer of Applied and Computational Mathematics Seminar (2009/8 – 2010/8)
- Editor for the Industrial Mathematics Institute preprint series (2005/4 – 2010/8)
- Editor for the Mathematics Undergraduate Research preprint series (2005/8 – 2010/8)
- Undergraduate Advisor (2005/8 – 2006/8)
- Coach for the Mathematical Contest in Modeling (2006/2 – 2008/8)
- Organizer of the SIAM Student Chapter Seminar (2006/8 – 2009/8)
- Doctoral Dissertation Committees of 19 Ph.D. students since 2004
- Judge for the Student Research Competition of SIAM Student Chapter at USC (2005/3)

COMMUNITY

- Board member of the SC East Point Academy (2015/8-2017/6)
- Judge for 2005 USC Region II Science and Engineering Fair (Math./Comput. Sci. Section)
- Judge for the 19th (2005/4), 20th (2006/1), 21st (2006/12), 22nd (2007/12), 23rd (2008/12), 24th (2010/1), 26th (2012/1), 28th (2014/1), 29th (2015/1), 30th (2016/1), 31st (2017/1), 33rd (2019/1), 34th (2020/1) USC High School Math Contest

PROFESSIONAL

- External research evaluator for 23 Promotion and/or Tenure cases at peer institutions

Promotion to Full Professor

- ☐ Dr. Bo Dong, Department of Mathematics, University of Massachusetts at Dartmouth, 2023
- ☐ Dr. Steve Wise, Department of Mathematics, University of Tennessee at Knoxville, 2016
- ☐ Dr. Yanlai Chen, Department of Mathematics, University of Massachusetts at Dartmouth, 2019 & 2020
- ☐ Dr. Guannan Zhang, Computational and Applied Mathematics Group (Promotion to Senior Research Staff), Oak Ridge National Laboratory, 2020
- ☐ Dr. Kening Wang, Department of Mathematics and Statistics, University of North Florida, 2022
- ☐ Dr. Alexander Labovsky, Department of Mathematical Sciences, Michigan Technological University, 2022
- ☐ Dr. Bo Dong, Department of Mathematics, University of Massachusetts at Dartmouth, 2023

Promotion to Associate Professor & Tenure

- ☐ Dr. Kening Wang, Department of Mathematics and Statistics, University of North Florida, 2011
 - ☐ Dr. Maria Emelianenko, Department of Mathematical Sciences, George Mason University, 2011
 - ☐ Dr. Yanzhi Zhang, Department of Mathematics and Statistics, Missouri University of Science and Technology, 2014
 - ☐ Dr. Leo Rebholz, Department of Mathematical Sciences, Clemson University, 2014
 - ☐ Dr. Yanping Ma, Department of Mathematics, Loyola Marymount University, 2016
 - ☐ Dr. Yanaxiang Zhao, Department of Mathematics, George Washington University, 2018
 - ☐ Dr. Ying Wang, Department of Mathematics, University of Oklahoma, 2018
 - ☐ Dr. Qingshan Chen, Department of Mathematical Sciences, Clemson University, 2018
 - ☐ Dr. Hans-Werner van Wyk, Department of Mathematics and Statistics, Auburn University, 2020
 - ☐ Dr. Wei Guo, Department of Mathematics and Statistics, Texas Tech University, 2022
 - ☐ Dr. Yuan Liu, Department of Mathematics, Statistics and Physics, Wichita State University, 2022
 - ☐ Dr. Nan Jiang, Department of Mathematics, University of Florida, 2022
 - ☐ Dr. Vu Thai Luan, Department of Mathematics and Statistics, Mississippi State University, 2023
 - ☐ Dr. Loic Cappanera, Department of Mathematics, University of Houston, 2024
 - ☐ Dr. Zhiguo Yang, School of Mathematics Sciences, Shanghai Jiao Tong University, 2026
- Reviewed proposals for NSF, DOE and some international agencies
 - Reviewed papers for over 50 journals and conference proceedings
 - Member of Hiring Committee of Institute of Mathematical Sciences at Renmin University of China (2013 – 2015)
 - Texas Tech University Mathematics Graduate Program Review, April 2025